WATER QUALITY EVALUATION AND TREND ANALYSIS IN SELECTED WATERSHEDS OF THE ATLANTIC REGION OF CANADA

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Abstract. Water quality indices (WQIs) have been developed to assess the suitability of water for a variety of uses. These indices reflect the status of water quality in lakes, streams, rivers, and reservoirs. The concept of WQIs is based on a comparison of the concentration of contaminants with the respective environmental standards. The number, frequency, and magnitude by which the environmental standards for specific variables are not met in a given time period are reflected in WQIs. Further, the water quality trend analysis predicts the behavior of the water quality parameters and overall water quality in the time domain. In this paper, the concept of WQI was applied to three selected watersheds of Atlantic region: the Mersey River, the Point Wolfe River, and the Dunk River sites. To have robust study, two different water quality indices are used: Canadian Water Quality Index (CWQI), and British Columbia Water Quality Index (BWQI). The complete study was conducted in two steps. The first step was to organize and process the data into a format compatible with WQI analysis. After processing the input data, the WQI was calculated. The second step outlined in the paper discusses detailed trend analysis using linear and quadratic models for all the three sites. As per the 25 years trend analysis, overall water quality for agriculture use observed an improving trend at all the three sites studied. Water quality for raw water used for drinking (prior to treatment) and aquatic uses has shown improving trend at Point Wolfe River. It is further observed that pH, SO_4, and NO_3 concentrations are improving at Dunk River, Mersey River, and Point Wolfe River sites. To ascertain the reliability and significance of the trend analysis, a detailed error analysis and parametric significance tests were also conducted It was observed that for most of the sites and water uses quadratic trend models were a better fit than the linear models.

Keywords: CCME, EMAN sites, guidelines and standards, trend analysis, water quality index

1. Introduction

High profile events such as contamination of public water supplies in the towns of Walkerton, Ontario and North Battleford, Saskatchewan and numerous boil water orders in other municipalities in Canada, have been stern reminders that we can't take a healthy environment for granted. Outcomes from these tragedies include the realization that there is no established national water quality monitoring program in Canada; current monitoring is temporally and spatially fragmented; monitoring of key issues and stressors is lacking; and there is inadequate use of data and information generated by monitoring activities. As a result, water quality managers



Environmental Monitoring and Assessment **88:** 221–242, 2003.
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are unable to provide a comprehensive, national picture of the status and trends of water quality in Canada.

Water is one of the most important natural resources to sustain life. Ascertaining its quality is very crucial before use for water drinking, agricultural, aquatic life, recreational, or industrial purposes. However, all available water bodies are not suitable for all different uses. Water quality indices (WQIs) have been developed to assess the suitability of water for a variety of uses. These indices reflect the status of water quality in lakes, streams, rivers, and reservoirs. The concept of WQIs is based on the comparison of the water quality parameter with respective regulatory standards. The number of variables with exceedances (failed test), frequency of exceedances, and magnitude of exceedances of regulatory standards for specific parameters (variables) are reflected in WQIs.

Considering the simplicity and scientific basis of WQI, it is expected that these indices will provide meaningful summaries of overall water quality and possibly trends. While appreciating the importance and usability of WQIs, it is important to understand the limitations of WQIs. The WQIs are not intended to replace a detailed analysis of environmental monitoring and modelling, nor should they be the sole tool for the management of water bodies. However, WQIs can be used to provide a broad overview of environmental performance that can be conveyed to the public in an easy to understand format. The many advantages of these indices include their ability to represent measurements of a variety of variables in a single number; the ability to combine various measurements with a variety of measurement units in a single metric; and the facilitation of communication of the results. On the other hand, there are limitations in the use of WQIs: the loss of information by combining several variables to a single index value; the sensitivity of the results to the formulation of the index; the loss of information on interactions between variables; and the lack of portability of the index to different ecosystems (Zandbergen and Hall, 1998).

In Canada, there were different types of performance measures to assess water quality prior to the development of the Canadian Water Quality Index (CWQI). In 1997, to provide a homogenous system, the Water Quality Guidelines Task Group of the Canadian Council of Ministers of the Environment (CCME) formed the Water Quality Index Technical Subcommittee to standardize protocols for CWQI calculations and use (applications). The objective of this subcommittee was to assess different approaches and to develop a unified water quality index that could be used in Canada for water quality assessment.

In this paper, the authors briefly describe different approaches of water quality assessment that were in practice and the approaches proposed by CCME. In a later section of the paper, the water quality of three watersheds is discussed using the unified CCME's proposed water quality index.

1.1. WATER QUALITY INDICES USED IN CANADA

1.1.1. *Centre St Laurent*

The WQI developed at Centre St Laurent (CSL) comprised of three variables (CCME, 1999):

$$\mathrm{WQI} = [\Sigma(\mathrm{A_i} \times \mathrm{F_i})]/\mathrm{n}, \tag{1}$$

where, $\mathrm{A_i}$ is the mean level of exceedance for parameter i for guideline i. When a parameter value exceeds a guideline for that variable, the ratio 'exceeding value/ guideline value' is calculated. These ratios are summed and then divided by the number of times they occur. $\mathrm{F_i}$ is the frequency of values that exceed a guideline for a parameter relative to the total number of values obtained for that parameter ($\mathrm{F_i} = \mathrm{F_{exceed}}/\mathrm{F_{total}}$ for parameter i). n is the number of parameters. Depending on the water use under consideration, CSL calculates different water quality index.

1.1.2. *Quebec Index*

The Quebec index is based on an approach originally developed in New Zealand (Smith, 1990). It is based on the minimum of a number of sub-indices, which are calculated for each of the water quality parameters measured (CCME, 1999):

$$\mathrm{WQI} = \mathrm{minimum}(\mathrm{Isub_1}, \mathrm{Isub_2}, \ldots, \mathrm{Isub_n}). \tag{2}$$

This approach considers use of 'Delphi curves' for the calculation of the sub-indices. Delphi curves are based on expert opinions as to the significance of a particular level of a water quality component. These curves are non-linear and represent the aggregated opinion of a particular level of specific parameters in terms of the designated water use. The Quebec index represents the 'worst case impairment' of any of the variables measured (CCME, 1999).

1.1.3. *Alberta Index*

Alberta index uses a 'performance indicator' for water quality in their state of the environment reporting. It is represented as:

$$\mathrm{A} = (\mathrm{n_{exceed}}/\mathrm{n_{measure}}) * 100. \tag{3}$$

This is the percentage of the samples taken at a site where one or more of the variables exceed the objectives for a designated use. Common water uses in Alberta index include recreation, agriculture, and aquatic life.

1.1.4. *British Columbia Water Quality Index (BCWQI)*

The British Columbia water quality index (BCWQI) is based on a totally different approach that includes one extra factor not considered in other indices (CCME, 1999):

$$\mathrm{WQI} = (\mathrm{F_1^2} + \mathrm{F_2^2} + \mathrm{F_3^2})^{1/2}, \tag{4}$$

where: F_1 is the percentage of water quality guidelines exceeded; F_2 is the percentage of measurements in which one or more of the guidelines were exceeded; F_3 is the maximum (normalized to 100) by which any of the guidelines were exceeded.

The BCWQI (Rocchini and Swain, 1995) has the longest and most extensive development and application. Two of its factors are analogous to components of other indices: $\mathbf{F}_2$ is similar to the Alberta index, while $\mathbf{F}_3$ is similar to the Centre St Laurent index. $\mathbf{F}_1$ was not used in any of the other indices (CCME, 1999).

The British Columbia Water Quality Index (BCWQI) is based on a variety of objectives depending upon the designated water use: drinking, recreation, irrigation, livestock watering, wildlife, and aquatic life. Separate rankings were published based for each use. BCWQI actually uses an inverted scale, where an index value of 0–5 denotes excellent water quality while an index value 60–100 denotes poor water quality (Husain *et al.*, 1999).

The Province of Manitoba also supports the British Columbia approach. Based on its evaluation of four years of data on eight sites in Manitoba, it concluded that the British Columbia index appeared to give reasonable results for the Manitoba case.

1.1.5. *Ontario Index*

Ontario index uses BCWQI with modified F_3 so that it represents the average normalized exceedance rather than the maximum. It is done to avoid saturation problem in F_3 as noticed when large exceedances of objectives occurred (CCME, 1999).

1.1.6. *Other Water Quality Indices*

There are two main approaches for water quality index quantification - one based on absolute subindices, which are independent of the water quality standard, and other relative subindices, which depend on the water quality standard. Various indexing approaches discussed above come under the category of later approach (relative sub-indices).

In the context of absolute subindices, a variety of approaches have been proposed over the last two decades. All these approaches consider a quality variable to indicate the quality of the water on a zero (worst quality) to unity (best quality) scale. Among these approaches, Walski and Parker (1974) used an exponential function to represents the subindices of various quality variables. Landwehr (1979) suggested the Pearson type 3-distribution function to represents the subindices of all the quality variables. Bhargava (1985) modified the exponential function, but instead of yielding unity for zero pollutant concentration, his exponential function yields a subindex value as high as 8.8*E-06. Dinius (1987) used power function for the majority of subindices. Recently Swamee and Tyagi (2000) proposed aggregate index for water quality description.

1.1.7. *Canadian Water Quality Index (CWQI)*

1.7.1.1. *Conceptual Model of CWQI*

The CWQI has adopted the conceptual model of BCWQI (based on relative sub-indices). There are three factors in the index, each of which has been scaled between 0 and 100.

The values of the three measures of variance from selected objectives for water quality are combined to create a vector in an imaginary 'objective exceedance' space. 'Objectives' may refer to Canada-wide water quality guidelines or site-specific water quality objectives. The length of the vector is then scaled to range between 0 and 100, and subtracted from 100 to produce an index which is 0 or close to 0 for very poor water quality, and close to 100 for excellent water quality. Since the index is designed to measure water quality, it was felt that the index should produce higher numbers for better water quality. This earlier version was evaluated on synthetic data sets (Hart, 1998), and data sets from British Columbia (Phippen, 1998) and Newfoundland (Husain, 1998). These evaluations along with evaluations in Alberta and Ontario index revealed that significant problems arose due to the formulations for estimating frequency and amplitude.

As a result of these problems, the Alberta Agriculture Food and Rural Development in association with the Alberta Environmental Protection funded a research grant to study alternative formulations of the frequency and scope factors (Wright et al., 1998). The CWQI development subcommittee used this work as the basis for the final formulation of the CWQI. Strictly speaking, the drinking water use CWQI is intended for use with finished, treated municipal water quality as opposed to raw, untreated source water quality. The drinking water use CWQI described here is for raw untreated source water quality. It is done for general illustration and comparison purpose.

1.7.1.2. *Components of latest CWQI*

The revised CWQI consists of three factors:

Factor 1 (F1): Scope

This factor is called scope because it assesses the extent of water quality guideline non-compliance over the time period of interest. It has been adopted directly from the British Columbia Index:

$$F_1 = \left(\frac{\text{Number of failed variables}}{\text{Total number of variables}} \right) \times 100, \tag{5}$$

where **variables** indicate those water quality parameters with objectives which were tested during the time period for the index calculation.

Factor 2 (F2): Frequency
F_2 (**Frequency**) represents the percentage of individual tests that do not meet the objectives ('failed tests'):

$$F_2 = \left(\frac{\text{Number of failed tests}}{\text{Total number of tests}}\right) \times 100. \tag{6}$$

The formulation of this factor is drawn directly from the British Columbia Water Quality Index.

Factor 3 (F3): Amplitude
F_3 (**Amplitude**) represents the amount by which the failed test values do not meet their objectives, and is calculated in three steps:
(i) The number of times by which an individual concentration is greater than (or less than, when the objective is a minimum) the objective is termed an 'excursion' and is expressed as follows. When the test value must not exceed the objective:

$$excursion_i = \left(\frac{FailedTest\ Value_i}{Objective_j}\right) - 1. \tag{7}$$

For the cases in which the test value must not fall below the objective:

$$excursion_i = \left(\frac{Objective_j}{FailedTest\ Value_i}\right) - 1. \tag{8}$$

(ii) The collective amount by which individual tests are out of compliance is calculated by summing the excursions of individual tests from their objectives and dividing by the total number of tests (those which do and do not meet their objectives). This variable, referred to as the normalized sum of excursions, or *nse*, is calculated as:

$$nse = \frac{\sum_{i=1}^{n} excursion_i}{\#\ of\ tests} \tag{9}$$

(iii) F_3 is then calculated by an asymptotic function that scales the normalized sum of the excursions from objectives (*nse*) to yield a range between 0 and 100.

$$F_3 = \left(\frac{nse}{0.01nse + 0.01}\right). \tag{10}$$

The CWQI is finally calculated as:

$$CWQI = 100 - \left(\frac{\sqrt{{F_1}^2 + {F_2}^2 + {F_3}^2}}{1.732}\right). \tag{11}$$

The factor of 1.732 arises because each of the three individual index factors can range as high as 100. This means that the vector length can reach

$\sqrt{100^2 + 100^2 + 100^2} = \sqrt{30000} = 173.2$ as a maximum. Division by 1.732 brings the vector length down to 100 as a maximum.

We briefly present here application of the CWQI to three EMAN sites. For details, please refer (Husain, 2001).

2. Data Preparation

In this paper, the water quality index has been estimated for three different water uses: drinking, aquatic, and agriculture**.** For each site and water use, different sets of parameters (also knows as variables) have been used depending upon the availability of data and regulatory standards. The comprehensive list of all parameters used in the present study is given in Table I. Based on the preliminary evaluation of the data, the data processing was not straightforward due to the following variability and inconsistency in the raw data:

(1) A large number of parameters (variables) have been sampled.
(2) The majority of parameters (variables) have very few data.
(3) Several parameters (variables) have been analyzed by different methods over time and space.

Many parameters have been reported in more than one form (i.e., extract, dissolved, and total, etc.). Further analysis revealed that among the list reported, regulatory standards were not available for some of the parameters (variables). Thus, those parameters for which there are no available regulatory standards (e.g., dissolved and extractable trace metals) have been ignored in the WQI analysis. Furthermore, the data limitations (i.e, low record length) significantly impact the WQI analysis. In this study, only sets of data with a minimum of four parameters and sampled at least four times were used in the WQI analysis (Husain *et al.*, 1999). The use of statically large sample size ($n > 30$) further improves the accuracy and precision of the WQI analysis results. Certain water quality variables were analyzed for in significantly different ways over time; these results cannot be compared e.g. true color versus apparent color, NTU turbidity versus JTU turbidity.

3. Water Quality Analysis

3.1. DUNK RIVER, PEI (BEDEQUE BAY)

The summary statistics of the data are presented in Table II, while CWQI results are plotted in Figure 1. It is evident from Figure 1 that the water quality is ranked as ‘good’ for agricultural, ‘fair’ for drinking, and ‘poor’ for aquatic uses. The variations in the annual average CWQI for 10 years are shown in Figure 2. It is clear from CWQI results for drinking water use that the water quality consistently fluctuated around a mean value of 70. It has been further observed that since 1990

TABLE I
Water quality standards for different water uses

Water Quality Parameter	Unit	Water Uses		
		Drinking	Aquatic	Agriculture
Iron (Fe)	(mg/L)	0.3	0.3	5
Manganese (Mn)	(mg/L)	0.05	0.1^{BC}	0.2
Lead (Pb)	(mg/L)	0.01	0.001	0.2
Nickel (Ni)	(mg/L)	0.2^{BC}	0.025	0.02
Zinc (Zn)	(mg/L)	5	0.03	5
Aluminum (Al)	(mg/L)	0.2^{BC}	0.005	5
Chromium (Cr)	(mg/L)	0.05	0.002	0.008
Copper (Cu)	(mg/L)	1	0.002	1
Specific Conductance (Cond)	(uS/cm)	700^{BC}	N/A	N/A
Turbidity	(JTU)	1	N/A	N/A
Dissolved Oxygen (DO)	(mg/L)	N/A	5.5	NA
PH	Maximum	8.5	9	NA
	Minimum	6.5	6.5	NA
Phosphorous (P)	(mg/L)	0.1*	0.1*	NA
Dissolved Organic Carbon (DOC)	(mg/L)	5^{ON}	NA	NA
Arsenic (As)	(mg/L)	0.025	0.005	0.1
Selenium (Se)	(mg/L)	0.01	0.001	0.02
Mercury (Hg)	(mg/L)	0.0001	0.0001	0.003
Cadmium (Cd)	(mg/L)	0.005	0.00002	0.0051
Barium (Ba)	(mg/L)	1	NA	NA
Sodium (Na)	(mg/L)	200	0.062	NA
Total Nitrate (NO_3)	(mg/L)	10	0.062	10
Color	(real units)	15	NA	NA
Hardness	(mg/L)	500	NA	NA
Sulfate (SO_4)	(mg/L)	500	NA	1000
Chloride (Cl)	(mg/L)	250	NA	250
Floride (F)	(mg/L)	1.5	NA	1
Temperature	(°C)	15	NA	NA
Coliform	(No/Dl)	2.2	NA	1000

NA = Not Applicable or Not Available
BC British Columbia Provincial Water Quality Guidelines
ON Ontario Provincial Water Quality Guidelines
* Recommended as the Maximum Desirable Concentration
Unless otherwise specified data has been adopted from CCME Canadian Water Quality Guideline (1999)

TABLE II
Summary Statistics for Dunk River (data from 1966 to 1998)

Parameters	Count	Max	Min	Range	Mean	Median	Mode	Std dev	Skew	95 percentile
Fe, mg/L	247	4.2	0.001	4.199	0.1716	0.16	0.13	0.5792	4.4942	0.94
Mn, mg/L	248	0.95	0.01	0.94	0.0239	0.03	0.03	0.0676	10.359	0.12
Pb, mg/L	239	0.038	0.002	0.036	0.0011	0.002	0.002	0.0024	13.501	0.0021
Ni, mg/L	25	0.0027	0.0002	0.0025	1.7E-05	0.0002	0.0002	0.0005	4.7075	0.00062
Zn, mg/L	252	1	0.0005	0.9995	0.0081	0.01	0.01	0.0651	14.144	0.01725
Al, mg/L	234	21	0.016	20.984	0.3247	0.0685	0.042	1.9559	6.0778	5
Cr, mg/L	39	0.0074	0.0002	0.0072	4.2E-05	0.0002	0.0002	0.0011	5.8709	0.00104
Cu, mg/L	250	0.11	0.001	0.109	0.0025	0.002	0.002	0.0128	6.4130	0.01
Cond, uS/cm	445	293	18.8	274.2	198.598	212	239	39.058	−1.3648	244
Turb, JTU	439	700	0.1	699.9	7.0314	1.4	1.1	36.752	15.755	24
DO, mg/L	100	16	8.1	7.9	2.6820	12.2	11.8	1.3559	0.1242	14.81
PH	444	9	6.5	2.5	7.5207	7.8	7.9	0.3640	−0.7275	8.2
P, mg/L	246	0.96	0.005	0.955	0.0356	0.043	0.05	0.0960	5.8650	0.1975

TABLE II
Continued

Parameters	Count	Max	Min	Range	Mean	Median	Mode	Std dev	Skew	95 percentile
DOC, mg/L	318	15	0.5	14.5	0.9327	1.2	1	1.0791	7.6615	2.4
As, mg/L	118	0.005	0.0004	0.0046	0.0002	0.0007	0.0005	0.0005	5.0507	0.001405
Se, mg/L	39	0.0003	0.0001	0.0002	1.1E-05	0.0001	0.0001	5.7E-05	1.5694	0.00021
Ba, mg/L	39	0.176	0.0994	0.0766	0.0124	0.154	0.161	0.0201	−0.6239	0.171
NO_3, mg/L	357	6.5	0.08	6.42	2.2888	3	2	0.7596	0.2322	3.824
Color, Rel units	326	400	5	395	9.5458	5	5	29.389	8.4365	40
Hardness, mg/L	43	111	12	99	6.4246	71	67	16.803	−0.5840	89.36
Na, mg/L	327	12.2	1.4	10.8	4.2372	6	6.2	1.0546	1.399	6.9
SO_4, mg/L	334	18.8	2.1	16.7	4.8218	6.2	6	1.7179	2.070	9.2
Cl, mg/L	336	18	2.6	15.4	7.7383	10.45	10	1.7055	0.029	13.225
No/DL,	39	740	0	740	8.4061	33	56	165.1	2.870	373
Temp, °C	325	21.1	0	21.1	5.6353	7	1	6.1673	0.338	18
F, mg/L	200	0.53	0.04	0.49	0.0265	0.05	0.05	0.0509	8.065	0.1105

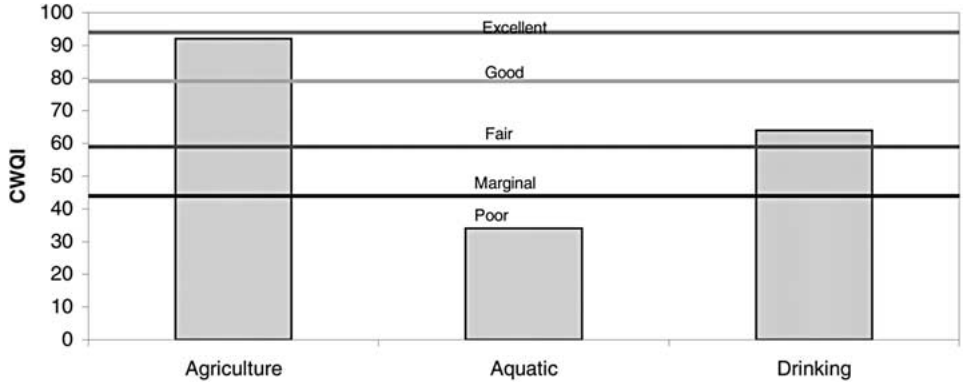


Figure 1. Canadian water quality index at Dunk River in PEI (Bedeque Bay EMAN Site) for entire period.

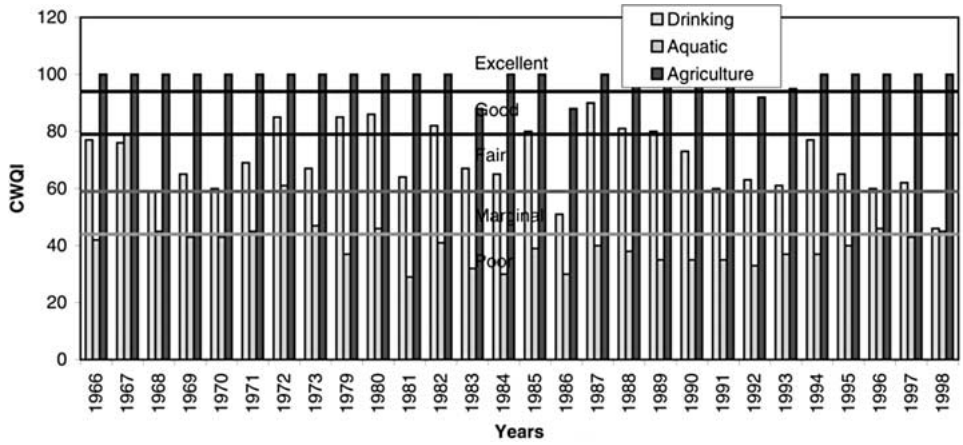


Figure 2. CWQI trend (drinking water) for Dunk River in PEI (Bedeque Bay EMAN Site).

values were almost constant with the exception of a sudden rise in 1994. In 1998 a sudden decrease in the water quality was observed. Unlike drinking water, the CWQI values for aquatic use show a relatively consistent trend. It is clear from Figure 2 that for the 1960s and 1970s, with the exception of 1967, CWQIs were constant. However, in the 1980s, the decreasing CWQI values reflected a lower quality; but more recently, the indices have increased. The CWQIs for agricultural use remained constant for most of the time.

3.2. Mersey River, NS (Kejimkujik National Park EMAN Site)

Data for the Kejimkujik National Park EMAN Site on the Mersey River was available from 1972 to 2000. The summary statistics of the data are presented in Table III. As listed in this table, the sample size varied for different parameters on a yearly basis. The CWQI results for all parameters (variables) for the entire time period

TABLE III
Summary Statistics for Mersey River in Nova Scotia (data from 1972 to 2000)

Parameters	Count	Max	Min	Range	Mean	Median	Mode	Std Dev	Skew	95 percentile
Fe, mg/L	1939	1.6	0.009	1.591	0.2160	0.2	0.16	0.0766	3.7537	0.34
Mn, mg/L	1947	0.12	0.01	0.11	0.0204	0.02	0.02	0.0073	3.5730	0.03
Pb, mg/L	193	0.012	0.001	0.011	0.0024	0.002	0.002	0.0014	4.6793	0.005
Zn, mg/L	184	0.29	0.002	0.288	0.0141	0.01	0.01	0.0264	7.9317	0.0294
Al, mg/L	1943	0.4	0.01	0.39	0.1799	0.17	0.14	0.0542	0.6912	0.28
Cu, mg/L	190	0.059	0.001	0.058	0.0027	0.002	0.002	0.0043	11.801	0.005
Cond, uS/cm	2804	316	6.2	309.8	35.196	33	31	11.896	12.947	44
Turb, JTU	1983	22	0.1	21.9	0.6891	0.6	0.6	0.5939	24.112	1.3
PH	2806	7	3.2	3.8	5.0106	5	4.9	0.3108	0.8167	5.5
P, mg/L	207	0.5	0.001	0.499	0.0133	0.008	0.005	0.0352	12.894	0.0284
DOC, mg/L	1757	28	0.5	27.5	9.3293	9	11	3.2357	0.6384	14
Cd, mg/L	192	0.02	0.001	0.019	0.0011	0.001	0.001	0.0013	13.482	0.001
NO_3, mg/L	1853	4.2	0.001	4.199	0.1386	0.11	0.1	0.1199	21.638	0.27
Color, Rel units	2489	320	5	315	84.314	80	100	25.443	0.5543	125
Na, mg/L	2489	15.7	0.3	15.4	3.4680	3.4	3.5	0.5273	5.9055	4.2
SO_4, mg/L	2438	21	0.5	20.5	2.9179	2.6	2.3	1.1709	5.56409	4.715
Cl, mg/L	2486	127	1.3	125.7	5.3486	5.2	5	2.6406	39.766	6.8
F, mg/L	264	0.1	0.02	0.08	0.0465	0.05	0.05	0.0137	0.49759	0.05

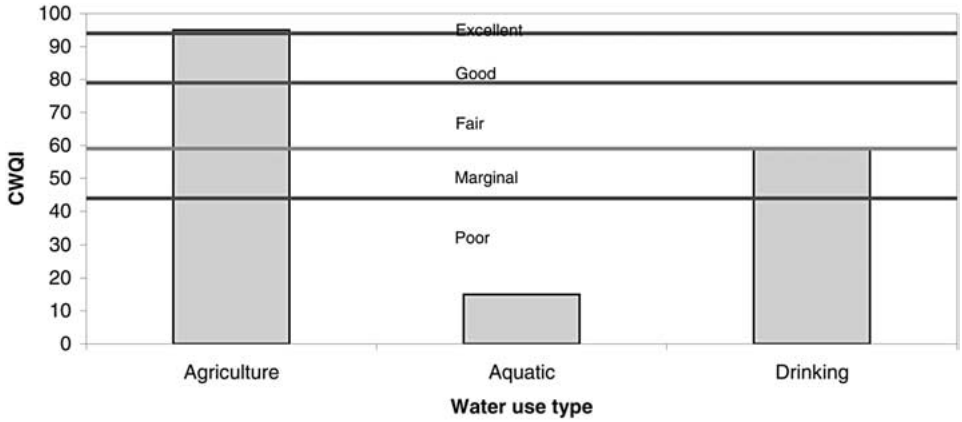


Figure 3. Canadian water quality index for Mersey River in Nova Scotia (Kejimkujik National Park EMAN Site) for entire period.

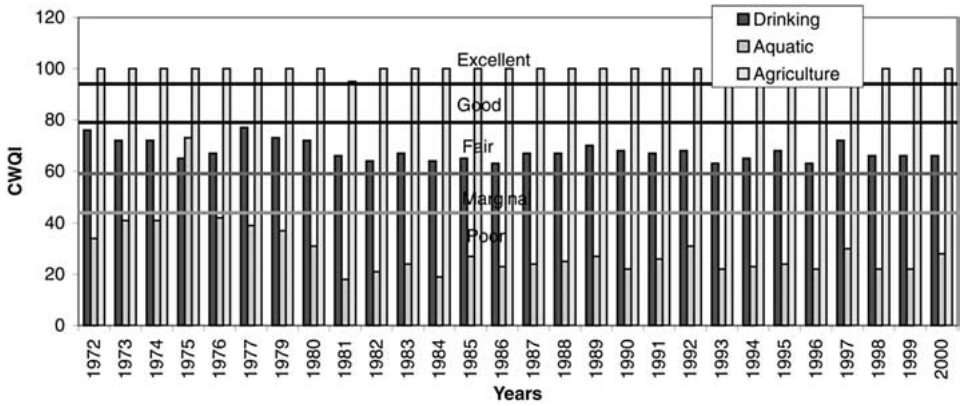


Figure 4. CWQI trend (drinking water) for Mersey River in Nova Scotia (Kejimkujik National Park EMAN Site).

are plotted in Figure 3. As shown in this figure, the water quality is rated 'good' for agricultural use, 'marginal' for drinking use, and 'poor' for aquatic use. The yearly variations in CWQI are shown in Figure 4. It is evident from Figure 4 that the drinking water quality is fairly constant with time. Figure 4 shows a variation with time in aquatic water quality indices. The time series analysis for aquatic use is similar to that for drinking water use but with slightly higher variations in the aquatic indices compared to the drinking water quality indices. The agricultural water quality indices are relatively constant for the duration of the study.

3.3. Point Wolfe River, NB (Fundy National Park EMAN Site)

Data for the Point Wolfe River EMAN Site was available from 1972 to 1999. The summary statistics of the data are presented in Table IV. As listed in this table, the

TABLE IV
Summary Statistics for Point Wolfe River (data from 1971 to 1999)

Parameters	Count	Max	Min	Mean	Median	Mode	Std Dev	Range	Skew	95 percentile
Fe, mg/L	257	0.43	0.002	0.0280	0.018	0.05	0.0417	0.428	6.3843	0.061
Mn, mg/L	257	0.07	0.00012	0.0102	0.01	0.01	0.0053	0.06988	7.6597	0.01
Pb, mg/L	270	0.018	0.0001	0.0020	0.002	0.002	0.0010	0.0179	12.690	0.002
Ni, mg/L	94	0.01	0.0001	0.0028	0.002	0.002	0.0028	0.0099	2.0234	0.01
Zn, mg/L	269	0.04	0.00074	0.0086	0.01	0.01	0.00408	0.03926	1.7094	0.01
Al, mg/L	251	0.7	0.00939	0.0660	0.05	0.04	0.0772	0.69061	5.7713	0.13
Cr, mg/L	37	0.0023	0.0001	0.0004	0.0002	0.0002	0.00052	0.0022	2.4614	0.0018
Cu, mg/L	269	0.0513	0.0002	0.0032	0.002	0.002	0.0053	0.0511	5.7387	0.01085
Cond, uS/cm	297	153	17.6	35.790	35	34	9.6174	135.4	6.1768	47.05
Turb,JTU	278	7.8	0.09	0.3301	0.2	0.2	0.5528	7.71	9.9406	0.7
DO, mg/L	40	14.7	9.2	11.872	11.8	12.2	1.5437	5.5	−0.0035	14.12
PH	297	7.5	5.1	6.75393	6.8	6.7	0.3067	2.4	−1.1142	7.2
P, mg/L	226	0.085	0.0009	0.0026	0.001	0.001	0.0059	0.0841	11.896	0.009

TABLE IV
Continued

Parameters	Count	Max	Min	Mean	Median	Mode	Std Dev	Range	Skew	95 percentile
DOC, mg/L	181	7.3	0.7	2.3596	2.1	1.6	1.1402	6.6	1.6768	5.2
As, mg/L	94	0.007	0.0001	0.0009	0.0005	0.0005	0.0014	0.0069	2.8209	0.005
Cd, mg/L	268	0.005	0.00001	0.0009	0.001	0.001	0.0003	0.00499	4.4438	0.001
Ba, mg/L	10	0.1	0.0127	0.0227	0.0143	0.0152	0.0271	0.0873	3.1549	0.06193
NO_3, mg/L	226	0.92	0.02	0.2893	0.245	0.09	0.1810	0.9	0.9103	0.62
Color, Rel units	297	70	0	10.006	5	5	8.1659	70	2.9141	25
Hg, ug/l	65	0.5	0.005	0.0444	0.02	0.02	0.0725	0.495	4.6848	0.186
Na, mg/L	297	17.3	0.37	2.3046	2.2	2.2	0.9679	16.93	12.597	2.9035
SO_4, mg/L	297	9.1	0.81	4.1419	4.2	4.3	0.8537	8.29	0.4691	5.3
Cl, mg/L	297	35	0.59	3.2110	2.9	2.9	2.0681	34.41	12.380	4.781
Temp, °C	244	23.5	0	7.2834	5.5	0	6.4157	23.5	0.5216	18.35
F, mg/L	176	0.1	0.02	0.0511	0.05	0.05	0.0099	0.08	3.5192	0.05

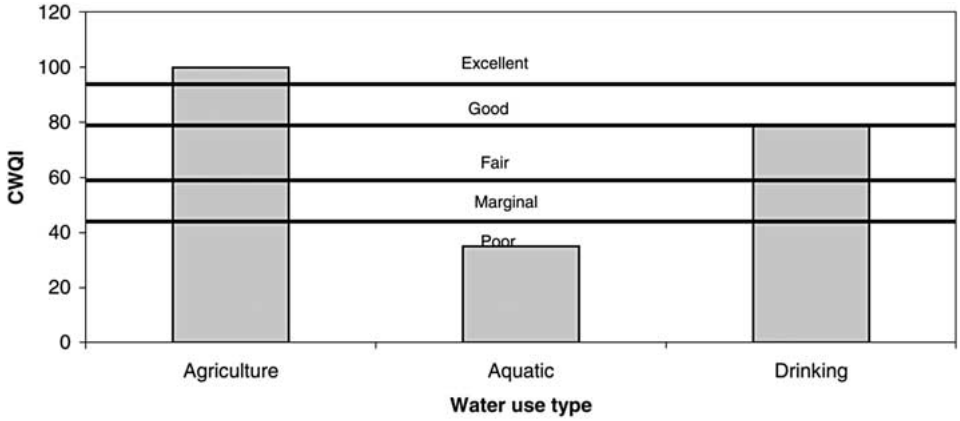


Figure 5. Canadian water quality index for Point Wolfe River in New Brunswick - Fundy National Park EMAN Site) for entire period.

data record length is not consistent with time, but reflects different sample sizes for a given parameter in a specified year. This also reflects a number of parameters (variables) measured in different years. The CWQI for the entire period is plotted in Figure 5. It is evident from this figure that the water in Point Wolfe River at the EMAN site is 'excellent' for agricultural use, 'fair' for drinking use, and 'poor' for aquatic use. An in-depth evaluation shows that the water at the Point Wolfe River Site is 'poor' for aquatic use due to relatively high concentrations of lead, cadmium, aluminum, and nitrate. The results of a time series analysis of CWQI for drinking water use are presented in Figure 6. It is evident from the results that water quality did not change much and remained constant from 1972 to 1999. Further, in the last three years there was an improving trend in water quality.

The results of the CWQI for aquatic use showed a rapidly changing trend (increasing and decreasing). Such inconsistency is due to the lack of sufficient observations. Between 1980 and 1999, there was a relatively long series of observations for each year and CWQI(s) during this period has shown a consistent increasing trend. The CWQI for agricultural use shows a fairly constant trend for the entire duration of the study.

4. Trend Analysis

4.1. DUNK RIVER, PEI (BEDEQUE BAY)

The trend plot for drinking water quality index is presented in Figure 7. As evident from the plot, the water quality index does not show any definite trend. However, the quadratic model shows a better fit with a decreasing trend in WQIs with time, but the fit seems statistically insignificant. The accuracy analysis of these trends is presented in Table V. Comparing the various errors, mean absolute percentage error

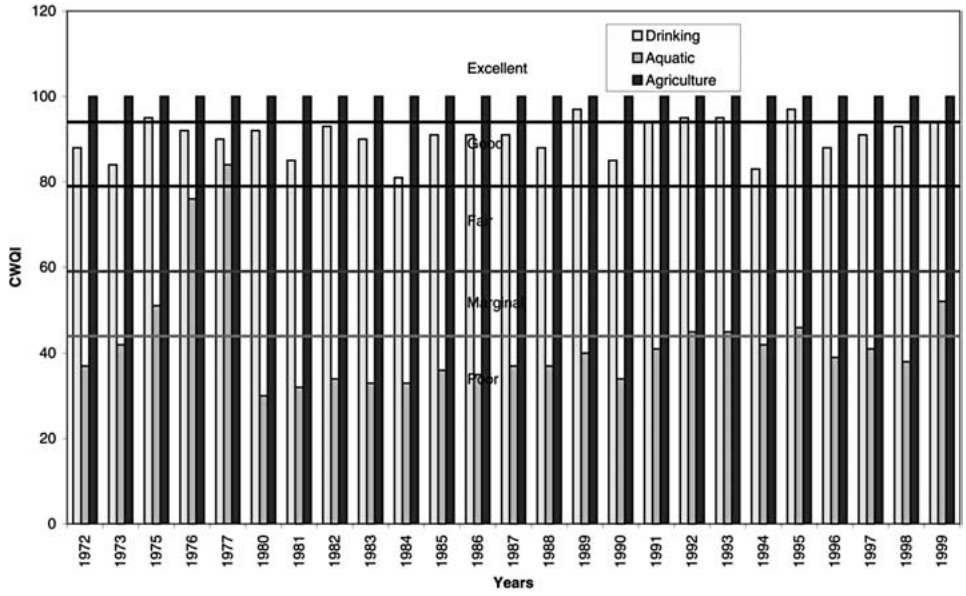


Figure 6. CWQI trend (drinking water) for Point Wolfe River in New Brunswick - Fundy National Park EMAN Site).

(MAPE), mean absolute deviation (MAD), and mean squared deviation (MSD) as listed in Table V with similar models for other sites, the error values are comparatively high and signify that the fit is not reliable. The results of the trend analysis for aquatic use also depict similar results. The trend for agriculture use shows a consistently uniform trend with excellent water quality.

4.2. MERSEY RIVER, NS (KEJIMUKUJIK NATIONAL PARK)

The result of the trend analysis for drinking water is presented in Figure 8. As shown in Figure 8, both linear and quadratic models were suitable for trend analysis. The quadratic model shows a steady decline in water quality with time, whereas the linear model does not show any definite trend with time. It is further observed that, for the quadratic model, the data is well within the acceptable 95% significance level. For the linear model, most of the data falls outside the range. Unlike drinking water CWQI, neither model fits well with the data in the case of the trend for aquatic water quality index. This may also be due to randomness and consistent fluctuation in the water quality data with time. Relatively speaking, the quadratic model is better than the linear one. An excellent predictive model fitting was found for the agricultural water trend as shown. An increasing trend predicted by the model was confirmed by water quality data and also by the parametric significance test.

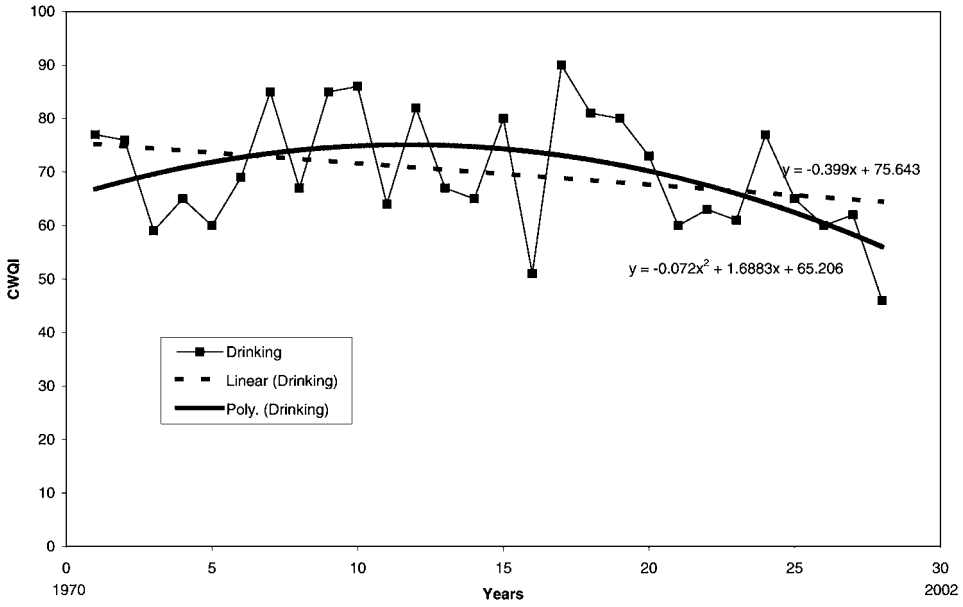


Figure 7. CWQI trend (drinking water) for Dunk River on PEI (Bedeque Bay EMAN Site).

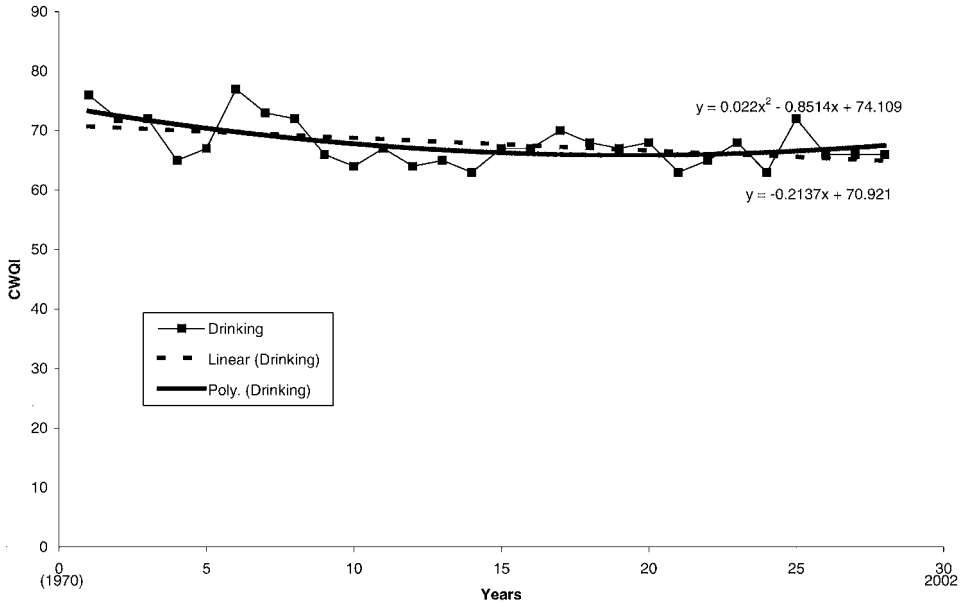


Figure 8. CWQI trend (drinking water) for Mersey River in Nova Scotia (Kejimkujik National Park EMAN Site).

TABLE V
Error analysis of trend model fitted to the water quality index at three EMAN sites

Parameters	Drinking		Aquatic		Agriculture	
	Linear	Quadratic	Linear	Quadratic	Linear	Quadratic
Dunk River (Bedeque Bay, PEI, EMAN Site)						
MAPE	13.24	14.67	11.86	2.42	2.42	2.42
MSD	8.98	6.14	5.18	2.29	2.29	2.32
MAD	110.8	81.26	56.79	11.56	11.56	10.55
Mersey River (Kejimkujik National Park, NS, EMAN Site)						
MAPE	4.01	3.71	20.16	15.86	0.35	9.01
MSD	2.73	2.53	5.79	4.79	0.34	0.37
MAD	11.02	9.38	81.46	64.01	0.83	27.4
Point Wolfe River (Fundy National Park, NB, EMAN Site)						
MAPE	3.716	3.74	18.88	17.10	0	0
MSD	3.310	3.33	8.47	7.76	0	0
MAD	17.07	17.04	150.7	134.6	0	0

TABLE VI
Water quality based on total length of records

Sites	Uses	Water quality
Dunk River (Bedeque Bay EMAN Site)	Drinking	Fair
	Aquatic	Poor
	Agriculture	Good
Mersey River (Kejimkujik National Park EMAN Site)	Drinking	Fair
	Aquatic	Poor
	Agriculture	Excellent
Point Wolfe River (Fundy National Park EMAN Site)	Drinking	Good
	Aquatic	Poor
	Agriculture	Excellent

4.3. POINT WOLFE RIVER, NB (FUNDY NATIONAL PARK)

The results of the trend analysis for the drinking water quality index for Point Wolfe River are presented in Figure 9. It is clear from this figure that the linear and quadratic models fit well and predict improvement in the water quality with

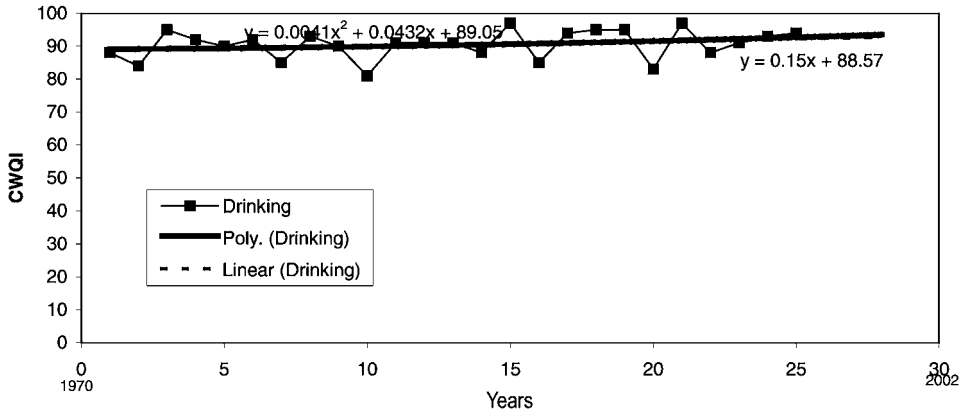


Figure 9. CWQI trend (drinking water) for Point Wolfe River in New Brunswick - Fundy National Park EMAN Site.

time. The parametric significance test further signifies that the fit is significant. In the case of the aquatic water quality index, neither models were statistically appropriate due to high (random) variability in water quality data. However, there is an improvement in the overall trend of water quality. The error analysis of the model (Table V) further suggests that the quadratic model's prediction is superior, though the accuracy is considerably lower than the fit of the drinking water quality index.

For agricultural use, CWQI, the results for linear and quadratic fits are well suited. This is confirmed by error analysis in Table V, which reveals that accuracy of the trend analysis for drinking water quality index is better than for the aquatic water quality index.

5. Summary and Conclusions

Water quality indices using the CWQI were calculated for aquatic, drinking, and agricultural uses. CWQI approach is based on three major factors: the number of times the objectives of environmental standards are not met; the frequency with which the objectives (Canada-wide, British Columbia or Ontario Canadian Water Quality Guidelines or site specific water quality objectives) are not met; and the amount by which the objectives are not met.

To apply the CWQI approach, the data from the three sites under study was screened and sorted to make it compatible with the requirements of the water quality analysis. The data was then arranged into complete data sets (considering all data for a site together), and time series data sets (annual data). CWQI has been computed for three water uses: drinking, aquatic, and agriculture. The CWQI study reveals that at all the three sites, the water is not of optimal quality for aquatic and drinking uses, whereas it is excellent for agricultural use. A summarized report

TABLE VII

Summary of parameter (variables) affecting the water quality with their number of exceedances

Parameters	Records	Drinking Numbers exceeding	Aquatic Numbers exceeding	Agriculture Numbers exceeding
Dunk River (Bedeque Bay EMAN Site)				
Ferrous	247	45	45	0
Manganese	248	34	15	3
Aluminium	238	49	234	0
Turbidity	439	290	–	–
Lead	246	26	239	0
Coliform	39	37	–	–
Nitrate	357	0	357	–
Mersey River (Kejimkujik National Park EMAN Site)				
Ferrous	1939	204	204	0
Aluminium	1934	576	1943	0
pH	2806	2794	2794	–
Lead	193	1	191	0
DOC	1757	1613	-	–
Nitrate	1853	0	1745	0
Point Wolfe River (Fundy National Park EMAN Site)				
Ferrous	259	2	2	0
Aluminium	251	6	251	0
pH	300	41	41	–
Lead	272	1	259	0
DOC	181	10	-	–
Nitrate	276	0	218	0
Cadmium	270	0	261	0

of the water quality for all sites and for three different water uses is presented in Table VI.

A time series analysis was also done using annual data sets, which indicates that there is a deteriorating trend in drinking water quality (prior to treatment) of two sites (Kejimkujik National Park, NS-EMAN Site and Bedeque Bay, PEI-EMAN Site). The limited number of records of the parameters (variables) significantly affected the time series analysis. A lower number of records generally tend to give a

higher WQI (suggesting a better water quality) because less data reduces the probability and proportion of exceedance with respect to the CCME Canadian Water Quality Guidelines and the frequency of exceedance. Important parameters (variables) affecting the water quality at these sites with their number of exceedances are listed in Table VII. It is evident from this analysis that total iron, total lead, pH and total cadmium are the parameters that frequently exceed the Guidelines.

References

Bhargava, D. S.: 1985, 'Expression for drinking water supply standards', *J. Environ. Engin., ASCE* **106**(4), 757–771.

CCME: 1999, Canadian Environmental Qualities Guidelines, Canadian Council of Ministers of the environment, Manitoba Statuary Publications, Winnipeg, Canada, Updated in 2000.

Dinius, S. H.: 1987, 'Design of an index of water quality' *Water Resource Bull.* **23**(5), 833.

Hart, W.: 1998, 'Alternative Formulations of Factors Used in the British Columbia Water Quality Index'. Report submitted to the Technical Subcommittee of the CCME Water Quality Guidelines Task Group.

Husain, T.: 1998, '*Application of National Water Quality Index Prototype in Newfoundland*', The State of Environment Reporting Task Group, The Canadian Council of Ministers of the Environment.

Husain, T., Khan, A. A. and Mukhtasor, A.: 1999, '*Final Report on Water Quality Index for Northwest Territories*', Water Management Planning Section, Yellowknife, pp. 12–13.

Husain, T.: 2001, 'Canadian Water Quality Index Determination for three EMAN Sites', Ecological Monitoring and Assessment Network (EMAN), Environment Canada, Burlington, ON.

Landwehr, J. M.: 1979, 'A statistical view of a class of water quality indices', *Water Resource Res.* **15**(2), 460–468.

Phippen, B.: 1998, 'Application of the National Water Quality Index Prototype in British Columbia'. BWP Consulting, 60 pp. June, 1998.

Rocchini, R., Swain, L. G.: 1995, 'The British Columbia Water Quality Index'. Water Quality Branch, Environmental Protection Department, British Columbia Ministry of Environment, Land and Parks. 13 pp.

Smith, D. G.: 1990, 'A Better Water Quality Indexing System for Rivers and Streams'. *Water Res.* **24**(10), 1237–1244.

Swamee, P. K. and Tyagi, A.: 2000, 'Describing water quality with aggregate index', *J. Environ. Engin., ASCE* **126**(5), 451–455.

Walski, T. M. and Parker, F. L.: 1974, 'Consumer's water quality index', *J. Environ. Engin., ASCE* **100**(3), 593–611.

Wright, C. R., Saffran, K. A., Anderson, A-M., Neilson, D., MacAlpine, N. and Cooke, S.: 1998, 'A Water Quality Index for Agricultural Streams in Alberta', 17 pp. + appendices.

Zandbergen, P.A. and Hall, K.J.: 1998, 'Analysis of the British Columbia Water Quality Index for Watershed Managers: A Case Study of Two Small Watersheds'. *Water Qual. Res. J. Can.* **33**(4), 519–549.